

PATRICK LIE

Software Engineer | AI & ML Development | Backend & Cloud | Mobile & IoT Systems
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SUMMARY

Fresh IT graduate (BSc Information Technology, IoT Specialization, Asia Pacific University, April 2026) who has built end-to-end systems across backend APIs, cloud deployment, mobile apps, IoT pipelines, and machine learning models. Recently expanded into LLM engineering, building and deploying a RAG chatbot and a fine-tuned language model, and into structured manual QA testing. Comfortable moving across different technical roles and picking up new tools quickly.

PROFESSIONAL EXPERIENCE

Software Engineer Intern (App Development) — Alva Production 2024 — Remote (WFH)

- Developed and maintained Flutter mobile apps, contributing to UI screens, reusable components, and feature implementation.
- Supported testing, debugging, and iterative improvements on an existing codebase to improve stability and UX.
- Assisted integration tasks and coordinated progress updates within an ongoing development roadmap.

Certificate of Achievement — With Distinction

Insurance Agent — Allianz 2022 – 2023

- Consulted clients to understand needs and communicate solutions clearly.
- Strengthened problem-solving and stakeholder communication through follow-ups and service support.

PROJECTS

TechDocs AI — RAG Chatbot for Technical Documentation 2026

Tech: Python, LangChain, ChromaDB, Groq (LLaMA 3.3 70B), Flask, Docker, Railway

- Built an end-to-end Retrieval-Augmented Generation (RAG) chatbot that answers questions from uploaded technical PDF documents.
- Implemented a document ingestion pipeline with semantic chunking and vector embeddings using ChromaDB.
- Deployed as a live REST API on Railway (asia-southeast1); production endpoint publicly accessible.

IndoIoT LLM — Fine-Tuned Language Model for Indonesian IoT 2026

Tech: Python, PyTorch, QLoRA/PEFT, Qwen2.5-3B, HuggingFace Transformers, Gradio

- Fine-tuned Qwen2.5-3B using QLoRA on 760 Indonesian IoT Q&A samples; achieved eval loss 0.52 with ~1% trainable parameters.
- Designed and curated the training dataset entirely from scratch using prompt engineering techniques.
- Deployed as an interactive demo on HuggingFace Spaces; tracked experiments with MLflow.

Freshly — Food Freshness Monitoring & Real-Time Spoilage Alerts (FYP, Solo) Completed

Tech: ESP32, Flutter, Firebase (Firestore/FCM), Cloud Functions, Google Compute Engine, Flask, Scikit-learn

- Built an end-to-end IoT system: ESP32 sensor hub → Firebase REST API → Flutter dashboard with real-time monitoring and alerts.
- Trained a Random Forest model to predict food spoilage risk, achieving high predictive accuracy on a ~12,000-sample dataset.

DIOT Urban Farming — Smart Monitoring & Automation Pipeline Completed

Tech: MQTT (EMQX), Node-RED, InfluxDB, Grafana, ESP32

- Designed a scalable IoT pipeline using MQTT for messaging and Node-RED for orchestration and automation.
- Implemented threshold-based monitoring logic, alerting, and data flow design for multi-node sensor deployments.

Chronic Disease Predictive Analytics — Patient Segmentation Completed

Tech: Python, Pandas, NumPy, Scikit-learn (K-Means, PCA), Matplotlib/Seaborn, PowerBI

- Implemented K-Means clustering to segment diabetes patient profiles using demographic and symptom features.
- Selected optimal cluster count using Elbow Method (WCSS) and Silhouette Score; produced PCA visualization and cluster profile tables.

QA Portfolio — Manual Testing (Web & Mobile) Apr – May 2026

Tech: Google Sheets, GitHub, Android Studio Emulator

- Executed 34 structured test cases across 3 projects: SauceDemo login page, e-commerce flow (web), and Sauce Labs Demo App (Android mobile).
- Identified and documented 2 bugs, including one Critical — a 100% reproducible app crash in the Android product catalog.

EDUCATION

Bachelor of Science (Hons) in Information Technology — Internet of Things 2022 – 2026

Asia Pacific University (APU), Malaysia — Graduated April 2026 | CGPA: 3.29 | Second Class Honours, First Division

SKILLS

Languages	Python, Dart, C, basic Java
AI / ML	LangChain, LlamaIndex, ChromaDB, FAISS, PyTorch, QLoRA/PEFT, HuggingFace Transformers, MLflow, Scikit-learn, TensorFlow/Keras, Pandas, NumPy
Backend & Cloud	Flask, FastAPI, REST APIs, Docker, Google Cloud (GCE, Cloud Functions, Cloud Run), Firebase (Firestore, FCM), Railway, HuggingFace Spaces
Mobile	Flutter, Firebase
IoT & Embedded	ESP32, Arduino IDE, MQTT, Node-RED, InfluxDB, Grafana
Tools	Git, VS Code, Figma, PowerBI, SQL
QA & Testing	Manual Testing, Test Case Design, Bug Reporting, Functional Testing, Exploratory Testing, Mobile Testing (Android), Google Sheets (test docs)

CERTIFICATIONS & LANGUAGES

- Certificate of Achievement — Alva Productions Software Engineer Internship (App Department), With Distinction
 - Cisco Networking Academy — Network Security (Course Completion) — May 2024
 - IELTS 6.5 — English Proficiency (Dec 2021)
- Languages: Indonesian (Native) | English (Proficient — IELTS 6.5) | Mandarin (Basic)